

Companion LXXIV: Fisher Information Geometry of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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April 2026

Abstract

Fisher information geometry provides a Riemannian structure on statistical manifolds, quantifying the sensitivity of probability distributions to underlying model parameters. Applied to persistence diagrams, this framework measures how topological features respond to changes in cosmological dynamics. Building on the Hilbert-space embeddings, kernel methods, and statistical tests developed in previous Companions, this work introduces the Fisher information metric for persistence diagrams in the Relaxation-Driven Cyclic Cosmology (RDCC). The relaxation mechanism modifies the scale-dependent evolution of the gravitational potential, inducing characteristic changes in the Fisher geometry of topological ensembles. The resulting information metric provides a powerful tool for parameter sensitivity, model discrimination, and information-theoretic analysis of weak-lensing topology.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. Previous Companions developed Hilbert-space embeddings, kernel PCA, and maximum mean discrepancy (MMD) as tools for analyzing and distinguishing the Relaxation-Driven Cyclic Cosmology (RDCC) from Λ CDM.

Fisher information geometry extends this framework by introducing a Riemannian metric on the space of probability distributions of persistence diagrams. This metric quantifies how sensitively the distribution responds to changes in cosmological parameters. In RDCC, the relaxation mechanism modifies the gravitational potential in a scale-dependent manner, producing characteristic signatures in the Fisher information of topological ensembles.

2 Fisher Information Geometry

Given a family of probability densities $p(x|\theta)$ depending on parameters $\theta = (\theta_1, \dots, \theta_d)$, the Fisher information matrix is

$$I_{ab}(\theta) = \mathbb{E} [\partial_{\theta_a} \log p(x|\theta) \partial_{\theta_b} \log p(x|\theta)]. \quad (1)$$

This defines a Riemannian metric on the parameter manifold.

For persistence diagrams, we consider the induced density in an RKHS via the embedding $\Phi(D)$:

$$p(D|\theta) \propto \exp \left(-\|\Phi(D) - \mu(\theta)\|_{\mathcal{H}}^2 \right),$$

where $\mu(\theta)$ is the mean embedding.

3 RDCC Modification of Persistence Diagram Distributions

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_{\kappa} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (2)$$

This modifies the distribution of persistence diagrams:

$$p_{\text{RDCC}}(D|\theta) = p(D|\theta) \left[1 - \chi_p \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right].$$

The mean embedding shifts as

$$\mu_{\text{RDCC}}(\theta) = \mu(\theta) \left[1 - \chi_{\mu} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right].$$

4 Fisher Information in RDCC

The RDCC Fisher information matrix becomes

$$I_{ab}^{\text{RDCC}} = I_{ab} \left[1 - \chi_{I,ab} \left(\frac{\ell}{\ell_{\text{rel}}} \right)^{\alpha} \right]. \quad (3)$$

RDCC induces:

- enhanced sensitivity to large-scale topological modes,
- suppressed sensitivity to small-scale modes,
- a characteristic turnover near ℓ_{rel} ,
- modified curvature of the information manifold,
- altered geodesic distances between topological distributions.

5 Information-Geometric Signatures of RDCC

The Fisher metric reveals how RDCC modifies the information content of topological features. Key signatures include:

- increased Fisher curvature for long-lived persistence features,
- reduced curvature for short-lived features,
- anisotropic deformation of the parameter manifold,
- enhanced distinguishability of RDCC from Λ CDM at large scales,
- modified geodesic flow of topological distributions across redshift.

These effects provide a direct information-theoretic probe of the RDCC gravitational field.

6 Conclusion

Fisher information geometry provides a powerful framework for analyzing the parameter sensitivity of persistence diagrams in the Relaxation-Driven Cyclic Cosmology. The relaxation mechanism modifies the Fisher metric in a scale-dependent manner, producing distinctive signatures in the information geometry of weak- lensing topology. These effects offer a direct, information-theoretic window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the foundation for future studies of Fisher kernels, parameter forecasts, and information geometry in cosmological topology.

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